MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
Model Question Paper
MG4VACCHE200 – BASIC ENVIRONMENTAL CHEMISTRY
(2024 ADMISSION ONWARDS)

Duration: 1.5 HRS **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries 1 marks

1. Which layer of atmosphere contains the ozone layer [K] [CO:1]
   A. Troposphere B. Stratosphere C. Mesosphere D. Thermosphere

2. Which of the following is a naturally occurring greenhouse gas [U] [CO:1]
   A. Carbon dioxide B. Sulfur Hexafluoride
   C. Perfluorocarbon D. HFC

3. Carbon Capture and Storage (CCS) primarily involves: [U] [CO:1]
   A. Reducing fossil fuel extraction
   B. Capturing CO_2 at emission sources and storing it underground
   C. Converting CO_2 into oxygen
   D. Removing methane from the atmosphere

4. The main anthropogenic source of carbon dioxide emissions is: [U] [CO:1]
   A. Volcanic eruptions B. Respiration
   C. Burning of fossil fuels D. Ocean evaporation

5. Which of the following is a major cause of water pollution? [U] [CO:2]
   A. Afforestation B. Industrial effluents
   C. Rainwater harvesting D. Soil conservation

6. Acid rain is primarily caused by emissions of: [K] [CO: 2]
   A. CO_2 and CH_4 B. SO_2 and NO_x
   C. O_3 and CO D. NH_3 and VOCs

7. Hydroelectric power generation is based on the conversion of: [U] [CO: 3]
   A. Thermal energy into electrical energy
   B. Chemical energy into electrical energy
   C. Kinetic and potential energy of water into electrical energy
   D. Nuclear energy into electrical energy

8. Which of the following is a first-generation biofuel? [U] / [CO:3]
   A. Algal biodiesel B. Cellulosic ethanol
   C. Corn-based ethanol D.Synthetic fuels

9. Which of the following is a major environmental impact of mining? [U] / [CO:3]
   A. Soil fertility improvement B. Acid mine drainage
   C. Increased groundwater recharge D. Carbon sequestration

[1x9 = 9]

Part B

Short Answer Questions

Answer any 5 questions

Each question carries 4 marks

10. What is CO_2 sequestration [U] [CO:1]

11. Define greenhouse gases. [K] [CO:1]

12. State two long-term consequences of water pollution. [U] [CO:2]

13. Mention two causes of ozone depletion. [U] [CO:2]

14. Define sustainable materials. [U] [CO: 3]

15. What is biodiesel? [K] [CO: 3]

16. Why is energy storage important for renewable energy systems? [U] [CO: 3]

[4x5 = 20]

Part C

Short Essay Questions

Answer any 3 questions

Each question carries 7 marks

17. Define global warming potential (GWP) and give one example of a gas with a GWP much greater than CO_2? [U] [CO:1]

18. Discuss the categories of water pollution, giving suitable examples. [U] [CO:2]

19. Explain the sources, types, and health effects of particulate air pollution. [U] [CO: 2]

20. Discuss hydrogen as a sustainable energy source, including its production, storage, and applications. [U] [CO: 3]

21. Describe the production and use of ethanol fuel from biomass. [U] [CO: 3]

[7x3 = 21]